

COMPUTER NETWORKING

2.1 Computer Technologies

The computer, an electronic device, was first invented as a programmable machine designed to sequentially and automatically perform a sequence of arithmetic or logical operations, which were generally termed *computing*. The very first device called a *computer* was of huge size, occupying the space of a room in a building. With the passage of time, the size has shrunk, but the power of computing has increased to a great extent. Today, computer technologies not only mean the technologies associated with formal computing machines, but also multifaceted technical innovations, equipment, methods, and applications. A wide range of technical devices in our daily life use computing power now. For example, a cell phone could also have a calculator that can perform advanced scientific calculations like a personal computer. Operating systems are also used in smart phones and similar devices. Even a watch, a car, a door locking system, an elevator, and many other everyday devices could have computing capability. Hence, today's computer technologies mean a vast range of technologies that use computing power for their daily operations.

When it comes to the notion of using actual computers in a system, major computer technologies include the conceptual subject areas: computer networking, software engineering, artificial intelligence, database management system, supply chain management system, wireless communications, signal processing, numerical computations, number theory, quantum computing, computer security, cryptography, bioinspired computing, computer architecture, pervasive computing, and other emerging technologies like cloud computing, future Internet (FI), Internet of things (IoT), near-field communication (NFC), and so on. In this chapter, we briefly look into the basic issues

of computer networking. Based on the structure of the networks, the strategies of dealing with worms should be different; hence, we need to know these or revise our understanding.

2.2 Network Topology

Network topology means the pattern or the model by which various elements of a network are interconnected. The elements could be computers, switches, routers, hubs, printers, phones, and other communications devices. For the interconnection, a wireless or wired medium could be used. Network topologies may be either physical or logical. Physical topology is defined by the physical design of a network, including the devices, location, and cable installation. Logical topology refers to how data are actually transmitted or circulated in a network. Physical topology could be understood by examining the structure or formation of the network; however, for understanding the logical topology, we need to understand which path the data follow while traveling through the network links. In other words, physical topology is the actual visible model of the network, and logical topology is the conceptual connectivity pattern of the communications devices.

In broad terms, mainly four topologies are recognized: (1) bus topology, (2) ring topology, (3) star topology, and (4) mesh topology. However, in this book, we note a wide range of topologies that should be clearly distinguished without resorting to grouping them under the banner of any of the four mentioned categories. Instead of using the term *hybrid topology*, we prefer to allow the readers to clearly identify some topologies [1] that should be studied distinctively in any form of computer networking. It should be noted that both physical and logical topologies are considered here, and we do not classify the mentioned terms under physical or logical as some of them contain both notions in the structure and in the operational method.

2.2.1 Point-to-Point Topology

When two endpoints are directly connected with a permanent link, it is called a point-to-point topology. Switched point-to-point topologies are the basic model of conventional telephony. Figure 2.1 shows a diagram of point-to-point topology.



Figure 2.1 Point-to-point topology.

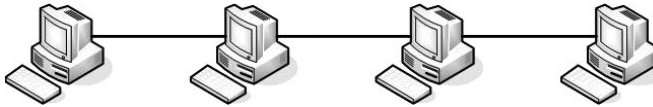


Figure 2.2 Daisy-chain topology.

2.2.2 Daisy-Chain Topology

Daisy chaining means connecting a device with another device in a sequence. In this topology, we basically extend the point-to-point topology by adding a number of devices (computers, switches, or other items) in a series. Figure 2.2 illustrates the daisy-chain topology. Such topology basically could take two shapes: linear and ring. Ring qualifies for being counted as a separate topology, which we present further in the chapter. In fact, many topologies are slight variants of other topologies, or the basic topologies are often present in the other complex topologies. Daisy chaining is also the basic method used for the bus topology (which is mentioned in the next section).

2.2.3 Bus (Point-to-Multipoint) Topology

When seen from the physical topology perspective, the bus topology is one of the major ones. A bus (or sometimes called a terminated bus) topology uses a single cable to which all of the nodes are connected. The cable is termed the *bus*. In the bus topology, as there is only one communication channel available, all the participating nodes compete with each other for the available bandwidth. In such a topology, a break of the cable at any point may disconnect the rest of the network. Figure 2.3 shows a bus topology.

2.2.4 Distributed Bus Topology

The distributed bus topology is a complex form of bus topology in which there is a main cable that connects several branch cables. It has

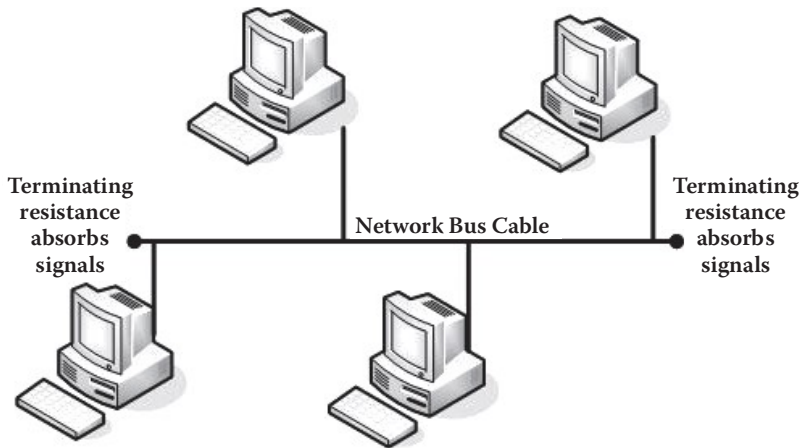


Figure 2.3 Bus topology.

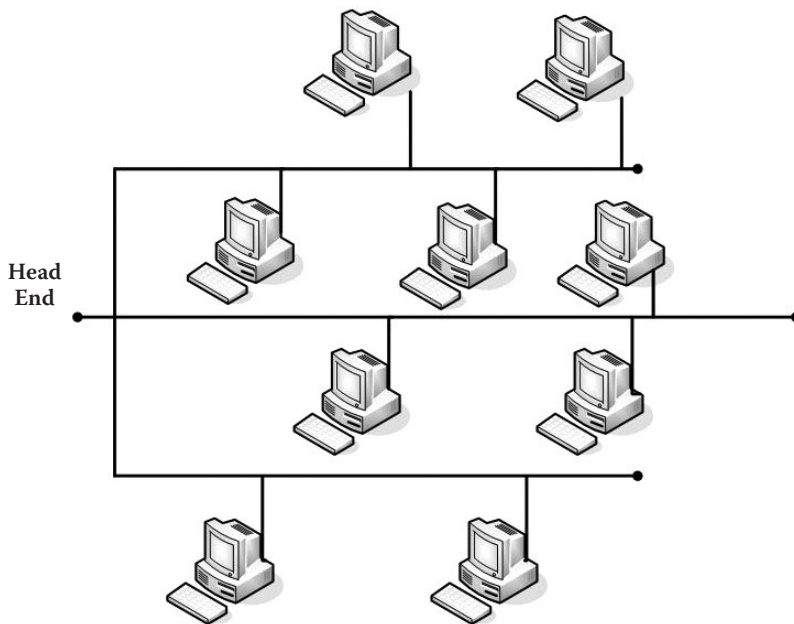


Figure 2.4 Distributed bus topology.

a head end, and each branch is terminated with absorber resistances. Figure 2.4 shows a diagram of such a topology. It should be noted that such a topology is susceptible to a single point of failure, and breaking the main cable disconnects the major portions of the network. In some scenarios, the topology could be useful.

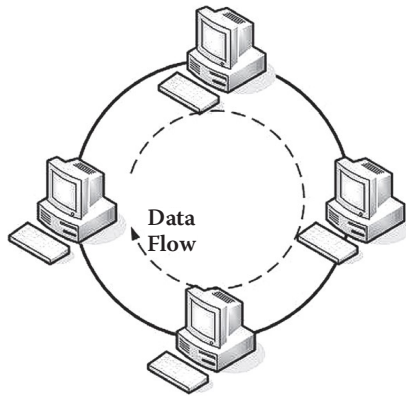


Figure 2.5 Ring topology.

2.2.5 Ring Topology

The ring topology uses daisy chaining of the devices in a circular fashion. The data within the topology flow circularly as shown in Figure 2.5. Each node in this topology checks a packet, and if it is not destined for itself, it forwards the packet to the next node. For a small-size network, this topology is suitable; in that case, it is relatively more robust than the bus topology.

2.2.6 Dual-Ring Topology

In a dual-ring topology, two concentric rings connect each node in a network instead of one network ring that is used in a ring topology. The redundant secondary ring in a dual-ring topology is used as a backup in case the primary ring fails. In these configurations, data move in opposite directions around the rings. Such a topology gives flexibility and robustness, so that each ring can operate independently of the other. When the primary one fails, the secondary ring is activated as shown in Figure 2.6.

2.2.7 Star Topology

In the star topology, there is a central entity that connects all the nodes with direct links. Figure 2.7 shows a diagram of the star topology. In fact, this is one of the topologies in which fault detection is the easiest. The number of node connections could be increased by increasing the connection links from the central entity. However,

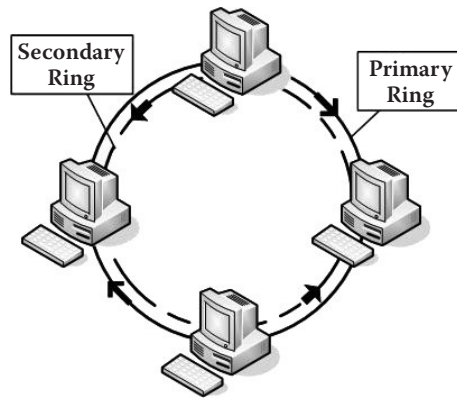


Figure 2.6 Dual-ring topology.

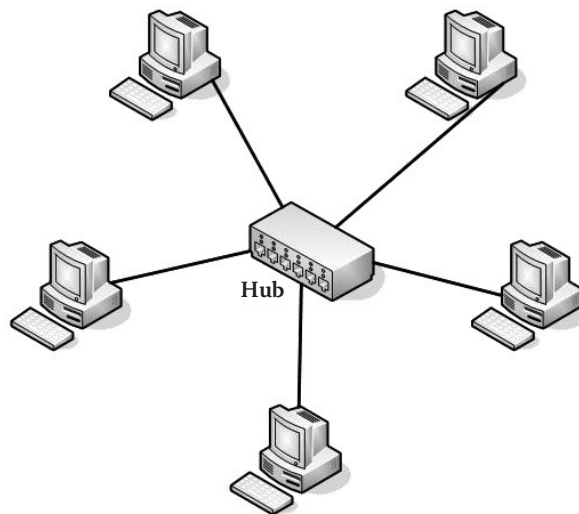


Figure 2.7 Star topology.

the scalability depends on the allowed limit of the link allowed with the central node. This topology is susceptible to a single point of failure as collapse of the hub (e.g., in Figure 2.7) breaks the entire network. However, a broken link with any node does not affect the link with another node, which is an advantage of this structure.

2.2.8 Star-Wired Bus Topology

In a star-wired bus topology, the nodes are connected with some kind of central entity in a star topology fashion; however, the difference is

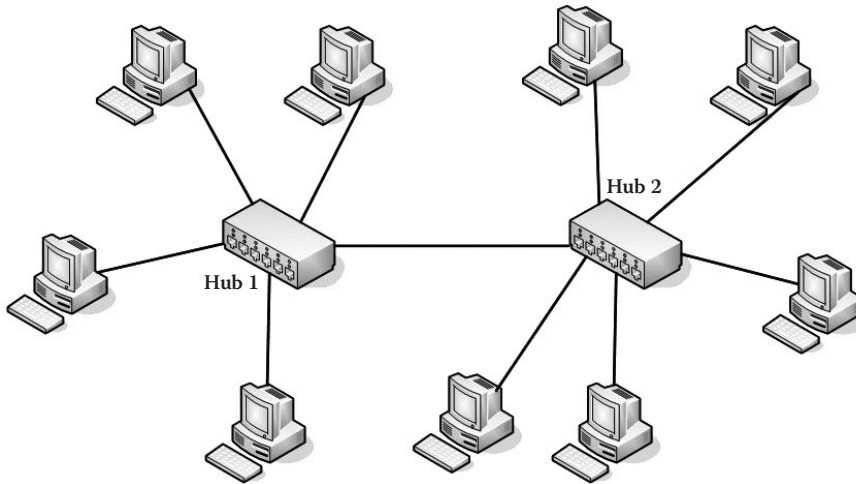


Figure 2.8 Star-wired bus topology.

that the central entities are again connected with each other. This is basically an extension of the star topology in a bus fashion to cover long distances. Figure 2.8 shows this topology.

2.2.9 Star-Wired Ring Topology

The star-wired ring topology physically looks like the star topology. However, the difference is that the data flow is in a circular fashion like that in a ring. Hence, the logical topology works like a ring topology. Figure 2.9 illustrates the star-wired ring topology; the dotted lines show the data flow through the central entity and the participating end nodes.

2.2.10 Mesh Topology

In a perfect mesh topology, each node is connected with every other node. This structure is robust and could sustain faults in multiple connections or links. Also, if one node goes down, other nodes could still communicate with multiple paths. A partial mesh topology has relatively fewer connections and a subset of a full mesh. Figure 2.10a and Figure 2.10b show the full-mesh and partial-mesh topologies, respectively.

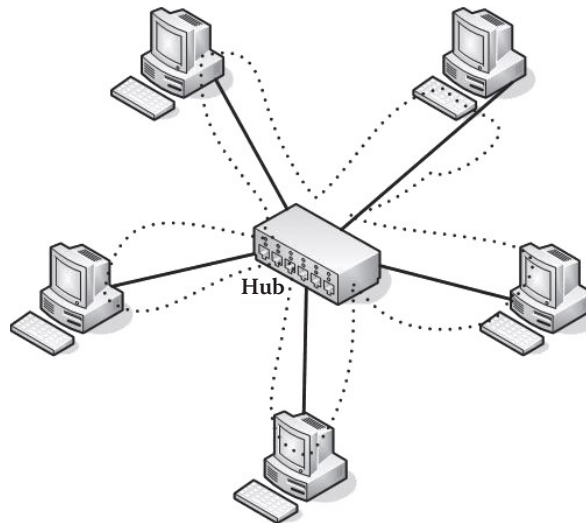


Figure 2.9 Star-wired ring topology.

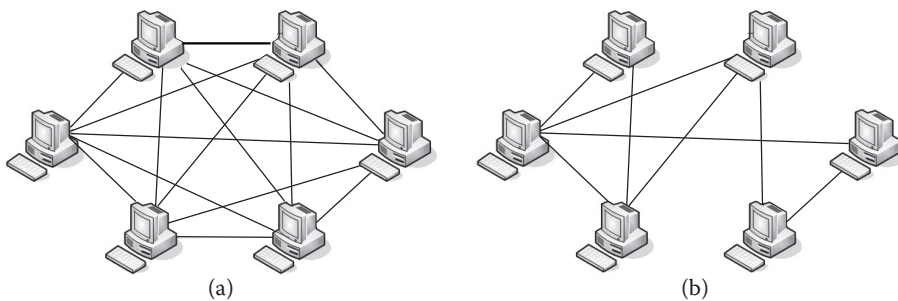


Figure 2.10 (a) Full-mesh topology; (b) partial-mesh topology.

2.2.11 Hierarchical or Tree Topology

In a hierarchical or tree topology, there are several levels of the connected components or the nodes. The higher node connects several nodes in the lower levels. A tree topology can be viewed as several star topologies connected in a hierarchical manner. Figure 2.11 shows a diagram of the tree topology. Of course, the number of connected nodes in the subsequent levels could be different, but overall the nodes have several layers where they exist, and the higher-level nodes support the lower-level nodes. The drawback of this topology is that if a higher-level node is down, the lower-level nodes depending on that higher-level node become disconnected from the main network.

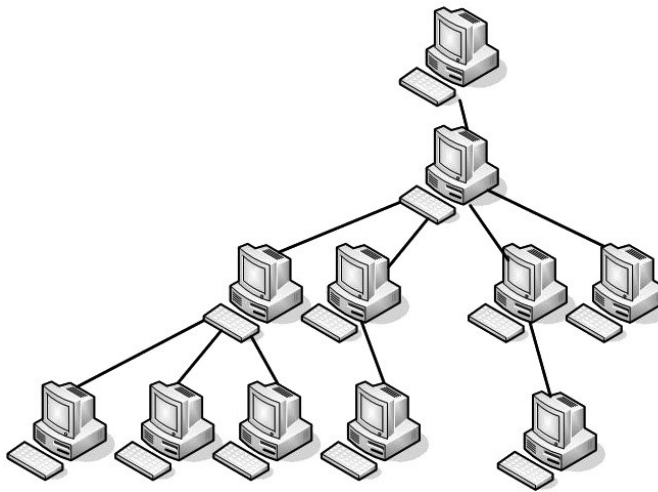


Figure 2.11 Hierarchical or tree topology.

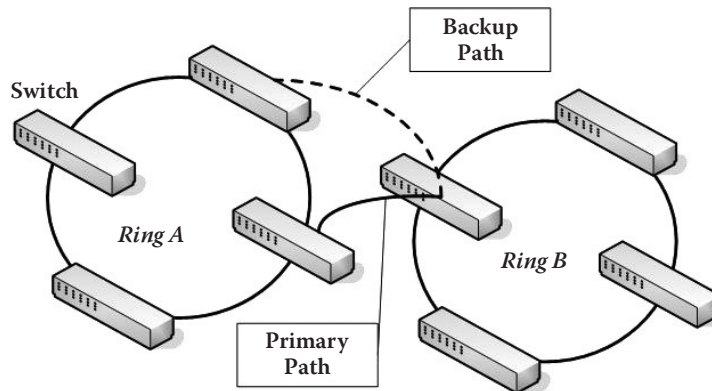


Figure 2.12 Dual-homing topology.

2.2.12 Dual-Homing Topology

The dual-homing topology adds reliability by allowing a device to be connected to the network by way of two independent connection points (points of attachment). One access point provides the operating connection; the other implements a standby or backup connection activated in the event of operating connection failure.

A dual-homing switch, with two attachments into the network, offers two independent media paths and two upstream switch connections. Loss of the link signal on the operating port connected upstream indicates a fault in that path, and traffic is quickly moved to the standby connection, accomplishing fault recovery. Figure 2.12 shows a diagram of the dual-homing topology.

After discussing various networking topologies, in the following sections we discuss some core definitions and terminologies that are essential for computer networking. Based on this knowledge, in further chapters we elaborate different aspects of dealing with worm technologies and their prevention mechanisms.

2.3 Internet Protocol

A protocol is the set of rules and principles for how the electronic devices communicate with each other by exchanging messages. Protocols may include various aspects like signaling, authentication, error detection, and error correction capabilities. The Internet Protocol, known in short as IP, is the protocol by which data are transmitted from one computing device to another over the Internet. Each computing device is termed a host or node, which is assigned at least one IP address that uniquely identifies it from all other computers or computing devices on the Internet.

The IP supports unique addressing for computers on a network. Most networks use the IP version 4 (IPv4) standard that features IP addresses 4 bytes (32 bits) in length. The newer IP version 6 (IPv6) standard features addresses 16 bytes (128 bits) in length. It should be noted here that IPv5 [2] was an experimental protocol based mostly on the OSI (Open System Interconnection) model that never materialized.

The IP is basically a connectionless protocol, which means that there is no continuous connection between the endpoints that are communicating. In this case, each traveling packet is treated as an independent unit of data without any relation to any other unit of data. The packets then are ordered the right way because of the TCP (Transmission Control Protocol), the connection-oriented protocol that keeps track of the packet sequence in a message. The TCP is discussed in the next section.

2.4 Transmission Control Protocol

The TCP is one of the core protocols of the IP Suite. TCP is one of the two original components of the suite, which complements the IP; therefore, the entire suite is commonly referred to as TCP/IP. TCP provides reliable, ordered delivery of a stream of bytes from a program

on one computer to another program on another computer. This is the protocol on which major Internet applications such as the World Wide Web (WWW), e-mail, remote administration, and file transfer rely. Other applications, which do not require reliable data stream service, may use the User Datagram Protocol (UDP), which provides a datagram service that emphasizes reduced latency over reliability.

2.5 IP Routers

A router is a device that is used to route packets based on their logical addresses (host-to-host addressing). It connects LANs (local-area networks) and WANs (wide-area networks). Each router has a routing table that helps for decision making about choosing routes for packets that it handles. A routing table could be dynamic, which allows updates using various routing protocols. Figure 2.13 shows the use of a router in an internetworking scenario.

2.6 Ethernet Switch

An Ethernet switch or network switch is used to connect multiple computers with each other. The connection with a switch looks similar to an Ethernet hub in terms of appearance as it connects a LAN, but it has some kind of intelligence involved in the mechanism. A switch not only receives data packets but also has the ability to inspect them before allowing them to pass to the next connected computer. So, a switch is able to figure out the source and the contents of the data and identify the appropriate destination. As a result of this uniqueness, it sends the data to the relevant connected system only, thereby using less bandwidth at high-performance rates.

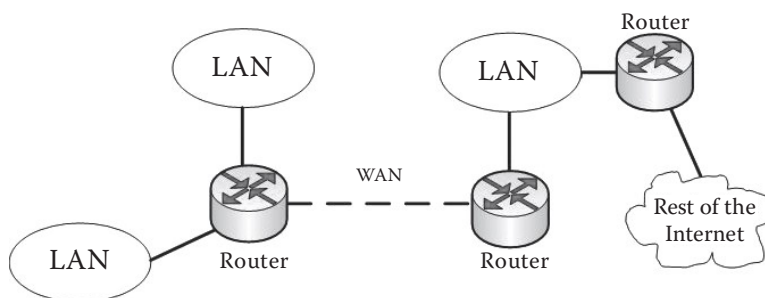


Figure 2.13 Router.

2.7 IP Routing and Routing Table

Internet Protocol routing is the process of selecting a path for network packets based on the destination IP addresses. Routing occurs at a sending TCP/IP host and at an IP router. In each case, the IP layer at the sending host or router must decide where to forward the packet. To make these decisions, the IP layer consults a routing table stored in the memory. Routing table entries are created by default when TCP/IP initializes, and entries can be added either manually or automatically.

A routing table or routing information base is a data table containing information about the routes to different network destinations (directly or indirectly), in some cases with distance information about the routes stored in the table. The router or any node using the TCP/IP has a routing table. Each table contains a series of default entries according to the configuration of the node, and additional entries can be added manually, for example, by administrators who use TCP/IP tools, or automatically, when nodes listen for routing information messages sent by routers [3].

When the IP forwards a packet, it extensively uses the routing table to determine

- The next-hop IP address
- Whether a packet is for a direct delivery, the next-hop IP address is the destination address in the IP packet, or whether it should be an indirect delivery (that is, the next-hop IP address is the IP address of a router)
- The next-hop interface
- That the interface identifies the physical or logical interface that forwards the packet.

A typical IP routing table entry includes the following fields:

- Destination address
- Prefix length corresponding to the address or range of addresses in the destination
- Next hop
- Interface

- Metric that indicates the cost of a route so that the IP can select the best possible route among many other routes to reach a particular destination; it may refer to the number of hops or links to reach the destination

2.8 Discussion on Router

As noted previously, routers basically direct network data messages, or packets, based on internal addresses and tables of routes, or known destinations that serve certain addresses. Directing data between portions of a network is the primary purpose of a router.

2.8.1 Access Mechanisms for Administrators

It is very important that a network administrator controls the access to a router to keep the networks safe from outside attacks. Here, we discuss this briefly. Some of the views and information were taken from Antoine, Bongiorno, and Borza [4]. So, for detailed knowledge, readers are encouraged to consult the official document.

There are two types of access:

- Local
- Remote

Local access usually involves a direct connection to a console port on the router with a dumb terminal or a laptop computer. Remote access typically involves allowing telnet or SNMP (Simple Network Management Protocol) connections to the router from some computer on the same subnet or a different subnet. It is recommended only to allow local access because during remote access, all telnet passwords or SNMP community strings are sent in the clear to the router. If an attacker can collect network traffic during remote access, then the attacker can capture passwords or community strings. However, there are some options if remote access is required. Two options are

1. Establishing a dedicated management network. The management network should include only identified administration hosts and a spare interface on each router. Figure 2.14 shows an example scenario.

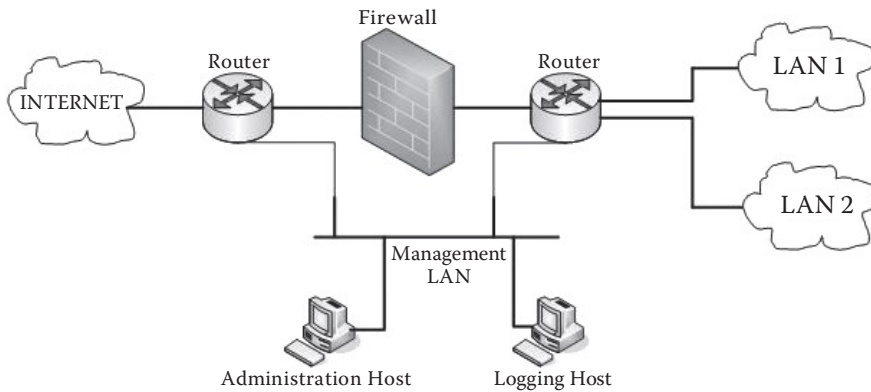


Figure 2.14 A sample management LAN (local-area network) for network administration.

2. Another method is to encrypt all traffic between the administrator's computer and the router.

In either case, packet filters can be configured to permit only the identified administration hosts management access to the router.

In addition to how administrators access the router, there may be a need to have more than one level of administrator or more than one administrative role. So, it is required to define clearly the capabilities of each level or role in the router security policy. For example, one role might be “network manager,” and administrators authorized to assume that role may be able to view and modify the configuration settings and interface parameters. Another role might be “operators”; administrators authorized to assume that role might be authorized only to clear connections and counters. In general, it is best to keep the number of fully privileged administrators to a minimum, maybe only one. In fact, too many cooks spoil the broth. The same applies in case of network administration.

2.8.2 Security Policy for a Router

There are several important tips to remember when creating the security policy for a router [4]:

- Specify security objectives, not particular commands or mechanisms. When the policy specifies the security results to be achieved, rather than a particular command or mechanism,

the policy is more portable across router software versions and between different kinds of routers.

- Specify policy for all the zones identified in Figure 2.14. Begin with physical security and work outward to security for the static configuration, the dynamic configuration, and traffic flow.
- Services and protocols that are not explicitly permitted should be denied. When representing the network policy in the router policy, concentrate on services and protocols that have been identified as explicitly needed for network operation; explicitly permit those and deny everything else.

In some cases, it may not be practical to identify and list all the services and protocols that the router will explicitly permit. A backbone router that must route traffic to many other networks cannot always enforce highly tailored policies on the traffic flowing through it because of performance concerns or differences in the security policies of the different networks served. In these types of cases, the policy should clearly state any limitations or restrictions that can be enforced. When drafting a policy, keep most of the directives and objectives high level; avoid specifying the particular mechanisms in the policy.

A security policy must be a living document. Make it part of the security practices of your organization to review the network security policy and the router security policy regularly. Update the router policy to reflect changes in the network policy or whenever the security objectives for the router change. It may be necessary to revise the router security policy whenever there is a major change in the network architecture or organizational structure of network administration. In particular, examine the router security policy and revise it as needed whenever any of the following events occur:

- New connections made between the local network and outside networks
- Major changes to administrative practices, procedures, or staff
- Major changes to the overall network security policy
- Deployment of substantial new capabilities (e.g., a new VPN [virtual private network]) or new network components (e.g., a new firewall)
- Detection of an attack or serious compromise

When the router security policy undergoes a revision, notify all individuals authorized to administer the router and all individuals authorized for physical access to it. Maintaining policy awareness is crucial for policy compliance.

Finally, some organizations have high-level policies that impose specific requirements on the contents of individual network security policies. Carefully check your router's security policy against any applicable high-level policy to ensure that it meets all the requirements.

2.8.3 Router Security Policy Checklist

The list that follows was designed in Antoine et al. [4] as an aid for creating router security policy. We cite the same here for readers. After drafting a policy, go through the list and check that each item is addressed in your policy.

2.8.3.1 Physical Security

- Designates who is authorized to install, deinstall, and move the router.
- Designates who is authorized to perform hardware maintenance and to change the physical configuration of the router.
- Designates who is authorized to make physical connections to the router.
- Defines controls on placement and use of the console and other direct-access port connections.
- Defines recovery procedures in the event of physical damage to the router or evidence of tampering with the router.

2.8.3.2 Static Configuration Security

- Designates who is authorized to log in directly to the router via the console or other direct-access port connections.
- Designates who is authorized to assume administrative privileges on the router.
- Defines procedures and practices for making changes to the router static configuration (e.g., logbook, change recording, review procedures).

- Defines the password policy for user/log-in passwords and for administrative or privilege passwords. Includes a list of conditions that require passwords to be changed (e.g., lifetime, staff changes, compromise).
- Designates who is authorized to log in to the router remotely.
- Designates protocols, procedures, and networks permitted for logging in to the router remotely.
- Defines the recovery procedures and identifies individuals responsible for recovery in the case of compromise of the router's static configuration.
- Defines the audit log policy for the router, including outlining log management practices and procedures and log review responsibilities.
- Designates procedures and limits on use of automated remote management and monitoring facilities (e.g., SNMP).
- Outlines response procedures or guidelines for detection of an attack against the router itself.
- Defines the management policy and update intervals for long-term secrets, such as those for routing protocols, NTP (Network Time Protocol), TACACS+ (Terminal Access Controller Access-Control System Plus), RADIUS (Remote Authentication Dial In User Service), and SNMP.
- Defines the key management policy for long-term cryptographic keys (if any).

2.8.3.3 Dynamic Configuration Security

- Identifies the dynamic configuration services permitted on the router and the networks permitted to access those services.
- Identifies the routing protocols to be used and the security features to be employed on each.
- Designates mechanisms and policies for setting or automating maintenance of the router's clock (e.g., manual setting, NTP).
- Identifies key agreement and cryptographic algorithms authorized for use in establishing VPN tunnels with other networks (if any).

2.8.3.4 *Network Service Security*

- Enumerates protocols, ports, and services to be permitted or filtered by the router, for each interface or connection (e.g. inbound and outbound), and identifies procedures and authorities for authorizing them.
- Describes security procedures and roles for interactions with external service providers and maintenance technicians.

2.8.3.5 *Compromise Response*

- Enumerates individuals or organizations to be notified in the event of a network compromise.
- Identifies relevant configuration information to be captured and retained.
- Defines response procedures, authorities, and objectives for response after a successful attack against the network, including provision for preserving evidence and for notification of law enforcement.

If such policy is employed, it is considered that the network security is given enough emphasis, and the network could be vulnerable to fewer potential commonly known attacks. This is because it is nearly impossible to keep a network free of any kind of attack. That is why we talk about minimizing security-related attacks rather than keeping them away completely.

2.9 Network Traffic Filtering

Network traffic filtering is a way of limiting the traffic by employing appropriate measures for checking which packets are allowed and which are not. This is a critical method used to enhance network security by using a different set of rules and criteria.

Here, we note two methods of traffic filtering: (1) packet filtering and (2) source routing.

2.9.1 *Packet Filtering*

Packet filtering is a method by which network packets are examined as they pass through routers or a firewall (see Figure 2.14) and it is determined whether the packets should pass or not. Packets may be filtered based on their protocol, sending or receiving port, sending or

receiving IP address, or the value of some status bits in the packet. Packet filtering could be of two types: static or dynamic.

- 1. Static Packet Filtering:** This method does not track the state of network packets and does not know whether a packet is the first, a middle packet, or the last packet. It does not need to know whether the traffic is associated with a response to a request or is the start of a request.
- 2. Dynamic Packet Filtering:** This method tracks the state of connections to employ a stricter form of security. It can identify if someone is trying to fool the firewall or router. Dynamic filtering is especially important when UDP traffic is allowed to pass. It can notify if traffic is associated with a response or request.

2.9.2 Source Routing

Source routing allows a sender of a packet to partially or completely specify the route that the packet takes through the network. On the other hand, in nonsource routing protocols, routers in the network determine the path based on the packet's destination. It is somewhat good that the packets in source routing contain header information describing the route they are to take to the destination. However, this type of routing is also a security concern when an attacker may gain access to a network without going through the firewall. So, by disabling source routing on the routers' traffic filtering mechanism, network security could be enhanced.

2.10 Tools Used for Traffic Filtering or Network Monitoring

Traffic filtering basically is a combination of various strategies and policies that are used for filtering network traffic. However, there are some tools that help for better handling and categorizing of network traffic. These are known as network monitoring tools, network analyzers, or network traffic-filtering tools. Some commonly used tools are Ethereal, EtherDetect, and Wireshark. There are many other tools that may be released soon or are available on the market. Searching the Internet may provide more knowledge about the latest tools and trends.

2.10.1 *Packet Capture*

Packet capture is a task of capturing network packets that pass through a network. Packet capturing could be used for network traffic analysis and network monitoring. Critical information about the packets could be obtained by capturing; hence, it is a very important task in network traffic filtering. A packet capture mechanism could be present in different network devices, like routers, switches, load balancers, servers, and other network equipment. Packet capture could be used in different ways, as mentioned next.

2.10.1.1 Complete Packet Capture Complete packet capture means capturing all packets that cross a network segment, regardless of source, protocol, or other distinguishing bits of data in the packet. Complete capture is the unrestricted, unfiltered, raw capture of all network packets.

2.10.1.2 Filtered Capture Packet capture devices may have the ability to limit capture of packets by protocol, IP address, MAC (media access control) address, and so on. With the application of filters, only complete packets that meet the criteria of the filter (header and payload) are captured, diverted, or stored from the raw capture of all network packets.

2.11 Concluding Remarks

The intent of this chapter was to set the base of discussion for our book's subject matter. As many of these are commonly known, we have given here a snapshot of various terms and scenarios that may have our solutions and algorithms applied to them. For the general readers of the topic, this chapter could provide an overview of the basic computer networking terminologies. For the experts, this should be considered just a reminder of what is known regarding the current status of computer networking.

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